RCM AI Platform — Presenter's Guide

Oct 9, 2026 · @Tarun

The one-minute pitch

The RCM AI Platform shows that a small, local, fine-tuned model (Qwen3-1.7B, 1.1 GB) becomes trustworthy for Revenue Cycle Management when it is wrapped in retrieval, deterministic SQL and validation, instead of being asked to answer alone.

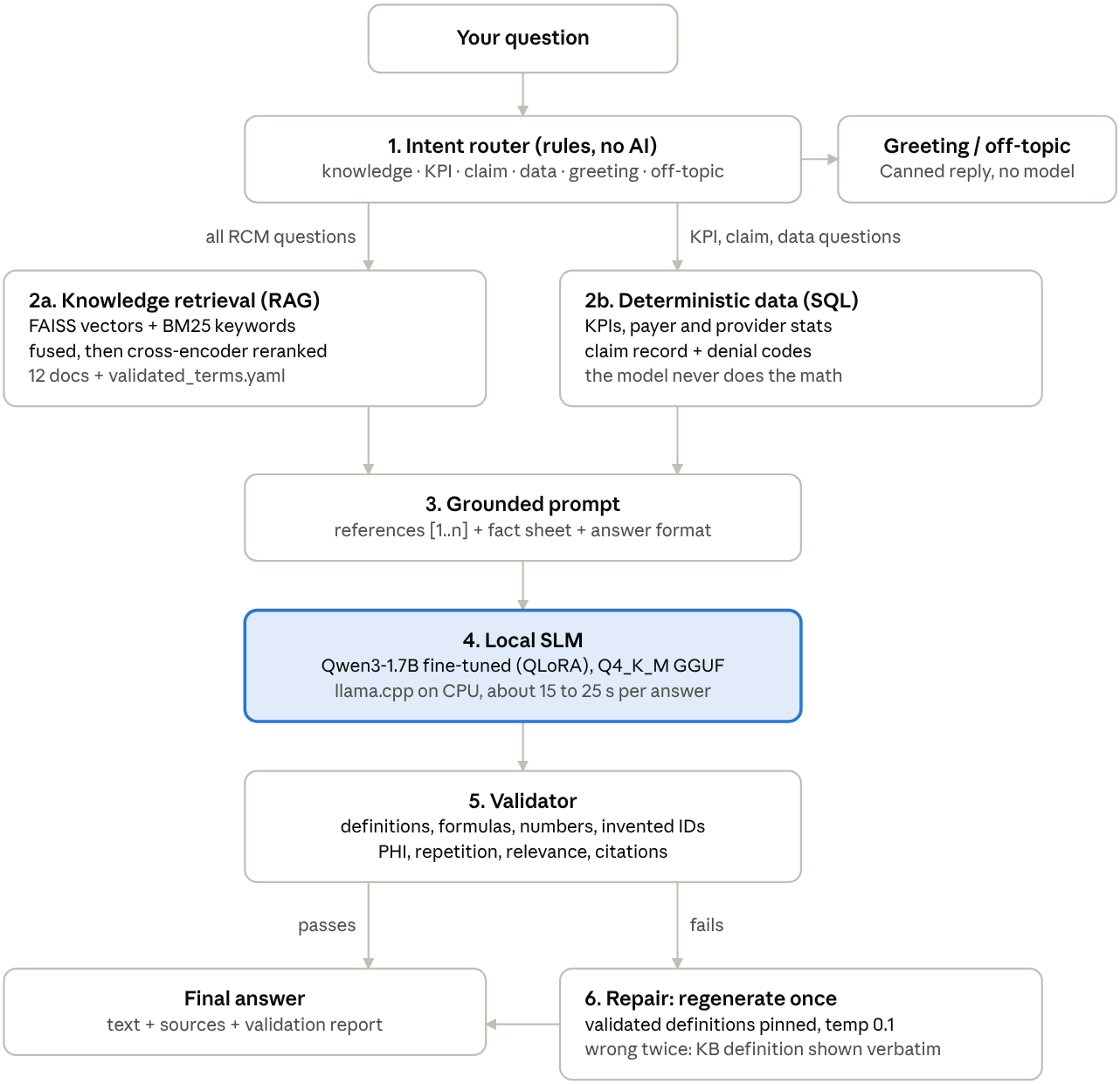
- **The model writes; it does not calculate or define.** Numbers come from SQL over the claims database. Definitions come from a curated knowledge base. The model's job is to explain them in plain language.
- **Everything runs locally.** The model is a 4-bit GGUF file run by llama.cpp on a laptop CPU. No claim data leaves the machine, and there is no per-token API cost.
- **Every answer is checked.** A validator compares the answer against the retrieved definitions, the SQL numbers and PHI patterns. A bad answer is regenerated with stronger context.
- **All data is synthetic.** Patients, claims and payers are generated; no real PHI exists anywhere in the system.

One line to open with: *"A 1.7-billion-parameter model on its own will invent definitions and do arithmetic badly. We never let it. SQL does the math, the knowledge base supplies the facts, and the model only explains — then a validator checks its work."*

Architecture: how one question flows

Every Chat and Copilot question passes through six steps. Only step 4 uses the AI model, and nothing it writes reaches the user without passing step 5.

The model only writes: retrieval and SQL feed it, a validator checks it



chat pipeline · backend/app/services/chat_service.py

The five-layer stack behind it: a Next.js + Tailwind + Recharts frontend, a FastAPI backend, llama.cpp running the GGUF model, a FAISS + BM25 knowledge index, and a SQLAlchemy database (SQLite locally, PostgreSQL/Neon via `DATABASE_URL`).

Step 1 — Intent router. A rule-based classifier picks the route. It is instant, explainable and cannot hallucinate a route, which a model-based router could.

Intent	Example question	What gets consulted
rcm_knowledge	"What is an ERA?"	Knowledge base only (top 4 passages)
kpi_analytics	"Which payer has the highest denial rate?"	SQL fact sheet + 2 knowledge passages
claim_specific	"Why was CLM-001234 denied?"	The claim record + passages chosen by its denial code
data_query	"List claims over $5,000"	Database
greeting / out_of_scope	"Hi", "Tell me a joke"	Nothing; fixed reply, no model call

Step 2a — Hybrid retrieval. Two searches run in parallel. FAISS finds passages with similar *meaning* (embeddings from `bge-small-en-v1.5`). BM25 finds passages with the same *exact words*, which matters for codes like CO-197. The two ranked lists are merged (reciprocal rank fusion), and a cross-encoder (`ms-marco-MiniLM-L-6-v2`) re-scores the 12 best candidates to keep the top 4.

Step 2b — Deterministic data. For data questions, Python/SQL computes the figures first and writes them into a "fact sheet". The model is told to use only those numbers.

Step 3 — Grounded prompt. References are numbered [1], [2]… so the answer can cite them. Knowledge answers use the format Answer / Why it matters / Example / Recommended action; data answers use Answer / Key drivers / Recommended actions.

Step 4 — Local SLM. Temperature 0.3, up to 512 output tokens, 4,096-token context. It runs on CPU, so an answer takes about 15–25 seconds; streaming shows words as they arrive.

Steps 5–6 — Validation and repair. If any *error*-level check fails, the model gets one retry with the validated definitions pinned to the top of the prompt and temperature lowered to 0.1. If it contradicts the knowledge base a second time on a definition question, the knowledge-base definition is shown verbatim and the user is told so. Invented claim IDs that survive the retry are masked, never shown.

Glossary: the RCM words you will say

These are the same definitions stored in `knowledge/validated_terms.yaml`, the file the validator checks the model against, so what you say on stage matches what the app enforces.

The claim lifecycle in one sentence: a patient is registered and their eligibility checked → the service is coded and a claim is submitted (837) → the clearinghouse/payer either *rejects* it at the front door or accepts it → the payer *adjudicates* it (pays, partly pays or *denies*) → the payer sends an *ERA* → the provider *posts* the payment → whatever is still unpaid sits in *AR* until it is collected, appealed or written off.

Term	Plain-English meaning	The one-line trap to avoid
ERA (Electronic Remittance Advice)	The payer's electronic explanation of payment to the provider (X12 835 file): billed, allowed, paid, patient share, with reason codes for every adjustment	It is not an EOB; it goes to the provider and is machine-readable
EOB (Explanation of Benefits)	A statement, mainly to the patient, of how the plan processed the claim	An EOB is not a bill
Claim rejection	The claim fails front-end edits (bad member ID, NPI, format) and never enters adjudication	Rejections are corrected and resubmitted, never appealed
Claim denial	The payer adjudicated the claim and refuses to pay all or part of it	Denials are appealed or corrected; they happen after adjudication
Adjudication	The payer's decision on a claim: pay, partly pay or deny	—
Billed amount	What the provider charged (list price)	Almost never what gets paid
Allowed amount	The most the payer's contract allows for the service	Billed − allowed = contractual adjustment

Term	Plain-English meaning	The one-line trap to avoid
Contractual adjustment	The discount the provider agreed to in its payer contract; written off, not owed by anyone	It is not lost revenue or a denial
Patient responsibility	Deductible, copay and coinsurance the patient owes	—
CARC / RARC	Standard codes on the ERA explaining why a line was paid differently (CARC) and extra detail (RARC). Group codes: CO = contractual, PR = patient, OA = other, PI = payer-initiated	e.g. CO-197 = no prior authorization
Payment posting	Recording payments, adjustments and denial codes against the right accounts, usually by auto-posting ERA files	It is not submitting claims
Prior authorization	Approval from the health plan *before* a service is delivered	Missing auth → denial (CARC 197 or 15)
AR (Accounts Receivable)	Money owed for services already billed but not yet paid, tracked in aging buckets (0–30, 31–60, 61–90, 91–120, 120+ days)	—
Write-off	A balance the provider stops trying to collect	—
COB (Coordination of Benefits)	Decides which plan pays first when a patient has two coverages	—
EFT	The actual bank deposit, matched to its ERA by trace number	The ERA explains the money; the EFT is the money

Term	Plain-English meaning	The one-line trap to avoid
Clearinghouse	The middleman that checks claims and routes them to payers	—

The eight KPI cards

In the default 90-day view, claim quality is healthy (95.6% clean, 84.8% first-pass) but cash is lagging: Days in AR rose 9.9 days to 53.9 and outstanding AR grew 36% to $652K. All eight numbers are computed in SQL by `backend/app/analytics/kpis.py`; the model never calculates them.

How to read a card. The big number is the selected period. The small coloured figure compares it with the *previous period of the same length*: green means it moved the good way, red the bad way, grey means neutral or flat. Rates and days change in **points** ("+0.5 pts", "+9.9 days"); money and counts change in **percent**. The sparkline shows the last 8 months. Hovering a card shows its exact formula.

Default period: the 90 days ending on the dataset's as-of date, 9 Jul – 6 Oct 2026, compared with 10 Apr – 8 Jul 2026. Claims are filtered by first-submission date.

Card	What it answers	Formula as implemented	Now (prev)	Good looks like
Total Claims	How much work went out the door?	Count of claims first submitted in the period	2,287 (2,157)	Neutral: volume, not quality
Clean Claim Rate	Are our claims built correctly?	Claims with no front-end rejection ÷ submitted claims × 100	95.6% (96.2%)	≥ 95%
Denial Rate	How often does the payer say no?	Adjudicated claims denied fully or partly ÷ adjudicated claims × 100	12.0% (12.6%)	< 5% best practice; 5–10% typical

Card	What it answers	Formula as implemented	Now (prev)	Good looks like
First Pass Resolution	How often is a claim paid with zero rework?	Adjudicated claims paid on first submission with no rejection or denial ÷ adjudicated claims × 100	84.8% (84.3%)	⩾ 90%
Days in AR	How many days of billing are sitting unpaid?	Outstanding AR at period end ÷ average daily billed charges over the trailing 90 days	53.9 (44.0)	< 40 days; top performers < 30
Collection Rate (net)	Of what we were allowed to collect, how much did we?	Payments ÷ (billed − contractual adjustments), closed claims only, × 100	92.1% (85.4%)	⩾ 95%
Total Revenue	How much cash came in?	Payer + patient payments received in the period, by payment date	$455.7K ($497.5K)	Up
Outstanding AR	How much are we still owed?	Unpaid balance of all claims at period end (a point-in-time snapshot)	$652.1K ($479.0K)	Down, or flat while volume grows

Why denial rate and first-pass rate are not mirror images. Denial rate counts *payer* refusals. First-pass resolution also fails a claim that was rejected at the front door and later paid, so FPR is always the stricter number.

Why clean claim rate is high but first-pass is lower. Clean claim measures the *front door* (format, IDs). First-pass measures the *outcome* (paid without rework). A perfectly formatted claim can still be denied for missing prior authorization.

Be ready for this one: "Why is Days in AR up 10 days?" AR grew 36% while billed charges grew 11%, so the numerator grew faster than the denominator. Part of that is timing: claims submitted in the last 30–60 days have not been paid yet, which also explains why

Payment Rate (below) is only 28%. Ask the Copilot "What is causing our AR to increase?" for the payer-level breakdown.

Six more KPIs appear on the Analytics page and in the API (`GET /api/kpis`):

KPI	Formula	Now (prev)
Total Billed	Sum of billed charges for claims in the period	$1.09M ($980K)
Total Denied Amount	Sum of denied dollars on claims in the period	$88.5K ($145.9K)
AR > 90 Days	AR on claims older than 90 days ÷ outstanding AR × 100	26.7% (27.7%)
Average Claim Value	Total billed ÷ number of claims	$475.87 ($454.44)
Payment Rate	Collected to date on this period's claims ÷ their billed charges × 100	28.4% (47.2%)
Avg Days to Adjudicate	Average days from submission to the payer's decision	25.1 (24.2)

Payment Rate is low because it is measured against *billed* charges (list price, before contractual discounts) and because recent claims are still in flight. Net Collection Rate is the fairer measure of collection performance.

All figures are from the synthetic dataset as of 6 Oct 2026; they change if the database is re-seeded.

Dashboard charts, one by one

The dashboard tells one story: claim volume grew by about a third in a year while monthly cash stayed flat, so money piled up in AR — and one payer, Cobalt Mutual, drives a large share of it.

Controls that change every chart. Date presets (30D, 90D, 6M, 12M, Custom) set the KPI period. Payer, provider and status dropdowns filter every card, chart and table. The trend charts always show the 12 months up to the period end. The current month (Oct 2026) holds only 6 days of data, so its bars are short — that is not a drop.

1. Claims submitted — bar chart

- **Shows:** claims per month, counted by first-submission date.
- **Now:** 580 in Oct 2025, rising to 754–801 a month in Jul–Sep 2026 (about +35%).
- **Say:** "Volume is up a third, so every rate on this page is measured on a growing base."

2. Denial rate — line chart

- **Shows:** denied claims ÷ adjudicated claims, grouped by the month the claim was *submitted*. Up is bad.
- **Now:** between 8.8% and 14.7%; peaks in May (14.4%) and July 2026 (14.7%). The 90-day figure, 12.0%, sits above the 5–10% industry band.
- **Caveat:** September looks better (9.9%) mostly because only 292 of its 798 claims have been decided yet. October has no decided claims, so the line stops.
- **Say:** "The jump from May onward lines up with Cobalt Mutual adding prior-authorization requirements on 1 May — ask the Copilot why our denial rate increased and it finds this from the data."

3. Payments received — stacked bars

- **Shows:** cash by the month it *arrived*, split into payer payments (bottom) and patient payments (top). Bar height = total cash.
- **Now:** $131K–$190K a month, peak $190K in March 2026; patients contribute roughly 12–19% of cash.
- **Say:** "Cash is flat around $150K a month while volume grew — that gap is where the AR growth comes from."

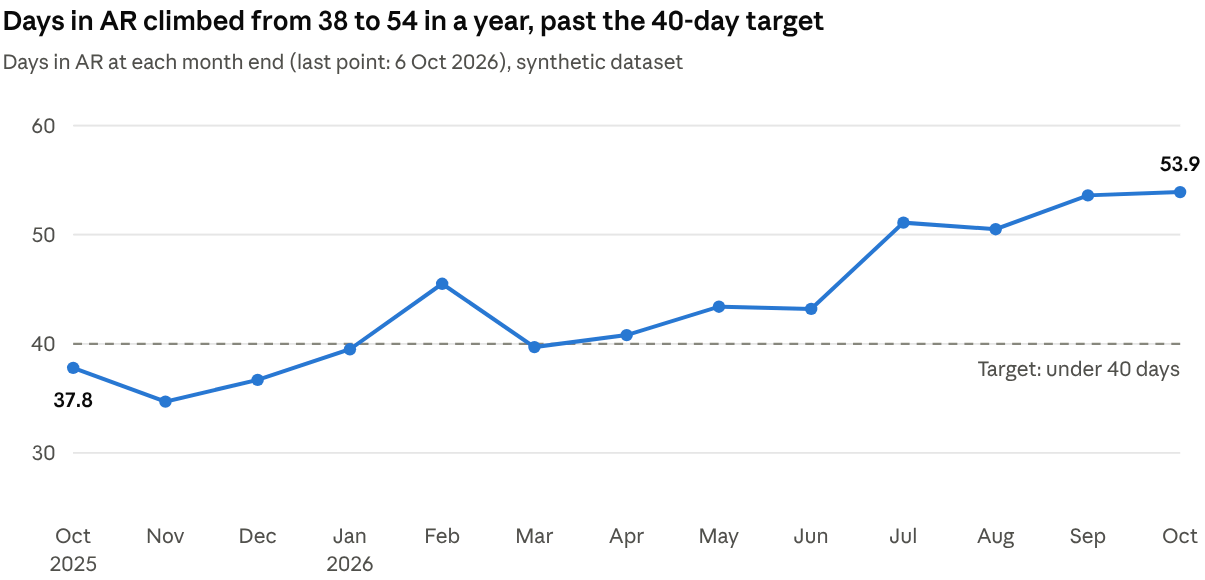
4. Clean claim & first-pass resolution — two lines

- **Shows:** front-end quality (clean claim, top line, about 95–98%) against outcome (first-pass resolution, lower line, 81–89%).
- **How to read:** the gap between the lines is claims that went out clean but still needed rework, mostly because of denials.
- **Now:** both dip in May 2026 (clean 94.6%, first-pass 81.3%) when front-end rejections jumped to 40.
- **Say:** "Our claims are well-formed; we lose money at adjudication, not at the front door."

5. Outstanding AR — area chart

- **Shows:** the total unpaid balance at each month end, a point-in-time snapshot rebuilt from transaction dates.
- **Now:** $378K in Nov 2025 → $652K on 6 Oct 2026. There is a one-off spike to $536K in Feb 2026 that recovers in March.
- **Say:** "AR has grown about 70% in a year, faster than volume."

6. Days in AR — line chart



GET /api/analytics/overview · ar_over_time, Oct 2025 – 6 Oct 2026

- **Shows:** outstanding AR ÷ average daily charges over the trailing 90 days — how many days of billing are sitting unpaid. Lower is better.
- **How to read it against chart 5:** if AR grows *because* volume grew, Days in AR stays flat. It rose here, so collections are genuinely slowing. The step up in July matches Cobalt Mutual slowing its decisions from 1 July (about 48 days instead of 24).
- **Say:** "We first broke the 40-day target in February, have stayed above it since April, and have been above 50 since July."

7. Top denial reasons — ranked bars

- **Shows:** denials in the period grouped by CARC code, longest bar = most denials. Hover a row for the code's description.

CARC	Meaning	Denials	Denied $	Preventable?
CO-197	Prior authorization absent	47	$43.3K	Yes
CO-97	Bundled into another service already paid	42	$9.2K	Yes
CO-50	Not medically necessary	41	$4.7K	No
CO-16	Missing information / billing error	18	$16.5K	Yes
CO-4	Procedure code inconsistent with modifier	8	$1.4K	Yes
CO-252	Attachment / documentation required	7	$3.8K	No

- **Say:** "About 70% of denials (115 of 163) are preventable. Prior authorization alone is 29% of denials and half the denied dollars — fix the auth check at scheduling first."

8. Claim status distribution — ranked bars

- **Shows:** each period claim by its *current* status, with its share and open balance.
- **Now:** Paid 1,378 (60.3%) · Pending 712 (31.1%, $337K open) · Denied 96 (4.2%, $62.6K open) · Partially paid 53 · Appealed 37 · Rejected 11.
- **Why "Denied" here (4.2%) is lower than the 12% denial rate:** this chart shows where each claim stands *today*. A claim that was denied and then appealed or paid counts in the denial rate but sits in another bucket here.
- **Say:** "Pending is the biggest open bucket and normal for recent claims; denied plus appealed — about $77K — is the money a team can act on now."

9. Payer performance — sortable table

- **Shows:** one row per payer for claims submitted in the period: claims, billed, collected, denial rate, clean claim, first pass, days to adjudicate, pending, open AR. Click a header to sort. The worst value in each "bad if high/low" column is flagged in red.
- **Now:** Cobalt Mutual is flagged on almost everything: denial rate 29.8% (about 3× every other payer), first pass 68.5%, 47 days to adjudicate, 223 pending and $131.5K open AR — 28% of the period's open AR from 18% of its claims.
- **Say:** "One payer explains most of the AR problem. That is a payer-relations conversation, not a billing-team problem."

Analytics page

The Analytics page answers "who" behind the dashboard: the same metrics split by payer and by provider. It opens on 12 months (7 Oct 2025 – 6 Oct 2026) rather than 90 days, so its numbers differ from the dashboard's.

1. Payer / Provider performance table

- **Shows:** the same sortable table as the dashboard's chart 9. A toggle switches between payers and providers. The worst value in each column is flagged red.
- **Columns, in plain words:** Claims (volume) · Billed (charges) · Collected (cash received on those claims) · Denial rate · Clean claim · First pass · Days to adjudicate (how long the payer takes to decide) · Pending (claims still waiting) · Open AR (unpaid balance).

Payers over 12 months, sorted by denial rate:

Payer	Claims	Denial rate	First pass	Days to adjudicate	Open AR	Payment rate
Cobalt Mutual Insurance	1,503	17.0%	80.8%	28.7	$161.1K	45.2%
State Medicaid (Synthetic)	1,376	13.5%	79.9%	34.7	$91.4K	29.8%
Lakeshore Health Partners	1,169	12.2%	85.5%	28.3	$86.0K	40.4%
Northwind Health Plan	1,727	11.3%	86.4%	21.7	$109.9K	50.8%
Medicare Part B (Synthetic)	2,106	9.7%	87.6%	16.7	$110.0K	39.3%
Sterling Care Assurance	945	8.7%	89.4%	18.8	$42.0K	58.0%

- **Say:** "Cobalt is 17% over the year but almost 30% in the last 90 days — it is getting worse, not just bad." (Switch the date preset from 12M to 90D live to show this.)
- **Second observation:** State Medicaid has the lowest clean claim rate (92.3% vs about 97% elsewhere) and the slowest decisions (34.7 days). Its problem is at the front door: eligibility and enrolment data.

Providers: Clearwater Imaging Center bills the most ($1.52M) and holds the most open AR ($202K). Summit Orthopedics and Lakeside Dermatology have the highest denial rates (15.9% each); Maple Grove Pediatrics the lowest (8.2%). Riverside Family Medicine submits the most claims (1,729) but bills only $290K — primary care means many small claims.

2. Cash collected — stacked bars

The same chart as dashboard chart 3 (payer + patient cash by payment month), filtered by this page's payer/provider selection.

3. Outstanding AR — area chart, and 4. Days in AR — line chart

The same as dashboard charts 5 and 6. Their value here is the filter: pick Cobalt Mutual in the payer dropdown and watch how much of the AR curve it carries.

5. Payment rate by payer (or provider) — ranked bars

- **Shows:** collected ÷ billed for claims submitted in the period, highest first. The grey sub-label reads "$X of $Y".
- **Now:** Sterling 58.0% · Northwind 50.8% · Cobalt 45.2% · Lakeshore 40.4% · Medicare 39.3% · Medicaid 29.8%.
- **How not to misread it:** it is measured against *billed* charges, so a payer with low contracted rates (Medicare, Medicaid fee schedules) scores low even when it pays every claim correctly. Read it next to denial rate and days to adjudicate before calling a payer "bad".

Denials page

Over 12 months there were 969 denials worth $449K. 74% of them were preventable, and appeals win about 60% of the time, so the money is recoverable but costly to chase. The page opens on 12 months.

Four summary cards

Card	Value (12M)	Meaning
Denied claims	969	Count of denial records in the period
Denied amount	$449K	Dollars the payers refused
Preventable	74.1%	Denials whose CARC is tagged preventable (everything except medical necessity CO-50 and documentation CO-252)
Appeal overturn rate	59.5%	Appeals the payer reversed ÷ appeals decided (about 178 of 299)

Say: "Three in four denials should never have happened, and when we appeal we win six times in ten. Prevention is cheaper than appeals."

1. Denials per month — side-by-side bars

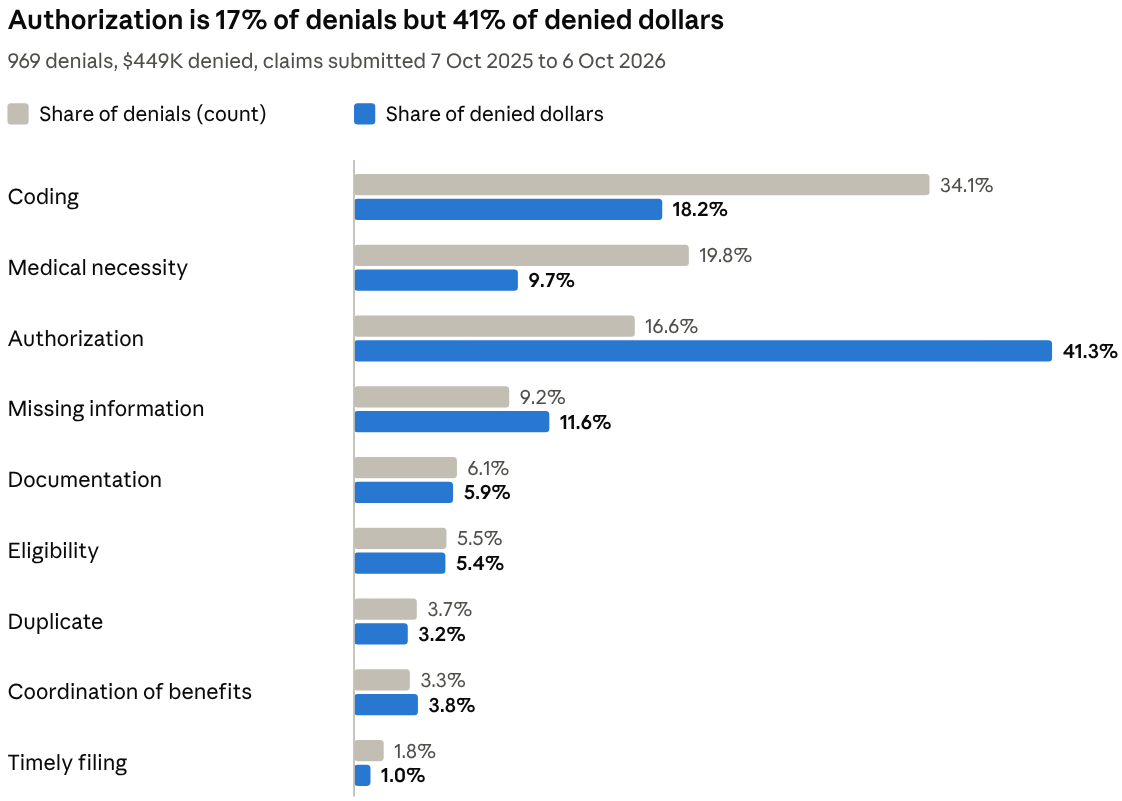
- **Shows:** for each submission month, all denials (first bar) next to the preventable ones (second bar). The gap between the two bars is the unpreventable part.
- **Now:** 71–100 denials a month; the preventable share stays around 70–80% every month, so this is a process issue, not a one-off.

2. Denial rate — line chart

The same line as dashboard chart 2, over 12 months.

3. Denial categories — ranked bars

- **Shows:** denials grouped into root-cause categories by count, with share and dollars in the sub-label.
- **The insight the app shows only in the sub-label — drawn here:**



GET /api/analytics/denials · categories, 12 months to 6 Oct 2026

- **Say:** "Coding is the most *frequent* problem, but authorization is the most *expensive* — only high-cost services need authorization (MRI, CT, echocardiograms, knee arthroscopy at $4,800), so each miss is expensive. Rank fixes by dollars, not by count."

4. Denial rate by payer — ranked bars

- **Shows:** each payer's denial rate, highest first; the top bar is highlighted. The sub-label reads "denied of adjudicated".
- **Now:** Cobalt Mutual 17.0% (217 of 1,279) · Medicaid 13.5% · Lakeshore 12.2% · Northwind 11.3% · Medicare 9.7% · Sterling 8.7%.

5. Denial reasons (CARC) — table

- **Shows:** one row per adjustment reason code: description, category, count, share, dollars, a "Preventable" tag, appeals decided and the overturn rate. Clicking a category opens the Claims page filtered to those claims.
- **Top rows:** CO-97 bundling 221 (22.8%) · CO-50 medical necessity 192 · CO-197 no authorization 161 ($186K, overturned 66% on appeal) · CO-16 missing information 89 · CO-4 modifier mismatch 69.

- **Say:** "CO-197 has the highest appeal win rate (66%) — the service was usually justified and the authorization was simply not on file. A front-desk auth check prevents both the denial and the appeal."

6. Denial rate by provider — ranked bars

- **Now:** Summit Orthopedics and Lakeside Dermatology are highest at 15.9%; Maple Grove Pediatrics is lowest at 8.2%.
- **Say:** "Two different root causes. Lakeside Dermatology has recurring modifier and bundling errors (CO-4, CO-97). Summit Orthopedics bills MRIs and knee arthroscopies, which Cobalt started requiring authorization for in May."

Claims, claim detail and "Analyze with AI"

The claim pages take the audience from the aggregate to a single claim. On one claim, the issues, actions and confidence come from rules, and the model only writes the narrative.

Claims explorer (the Claims page)

- A paged, sortable table of every synthetic claim. Columns: Claim ID, Submitted, Service date, Payer, Provider, CPT, Billed, Paid, Balance, Status, Denial code and Days in AR.
- Filters: the same date, payer, provider and status filters as the dashboard, plus a search box (claim ID, CPT, ICD-10 or CARC), a denial-category dropdown and an "Open balance only" switch.
- The Denials page links here: clicking a category in the CARC table opens this list already filtered.

Claim detail page — what each panel means

Panel	What it shows
Header	Claim ID and a coloured status badge
Amounts	Billed → Allowed → Payer paid, plus patient responsibility, patient paid, contractual adjustment, write-off and balance

Panel	What it shows
Service & coding	Service and submission dates, CPT, modifier, ICD-10 diagnosis, place of service, and authorization (number, "Required · none on file", or "Not required")
Payer & follow-up	Payer, provider, days in AR, and the timely-filing countdown ("X days left of Y" or "Passed N days ago")
Status history	Every status change with its date: submitted → accepted → adjudicated → denied/paid → appealed
Denials / Payments	The CARC-coded denial records and every payment posted

The amounts read as a waterfall: **billed − contractual adjustment = allowed; allowed = payer paid + patient responsibility; whatever is unpaid is the balance.** All patient data is synthetic: an opaque patient ID with only birth year, sex and state.

"Analyze with AI" — how the output is built

1. **Rules read the claim record** and produce the findings (each tagged high / medium / low / info), the recommended actions from a playbook per denial category, and a confidence score. Examples: "CARC 197: authorization absent" (high), "Timely filing risk" (high), "Awaiting adjudication" (medium if overdue), "Claim resolved" (info).
2. **Confidence is rule-based**, not the model's opinion: ≥ 0.85 = high, ≥ 0.70 = medium, below that = low. It drops, for example, when a claim is still pending or the filing deadline is uncertain.
3. **Retrieval uses the claim's own denial code and status**, not the question wording, so the knowledge passages match the actual situation.
4. **The model writes the narrative** (Claim summary / Current status / Potential issue / Explanation / Recommended next action) from that fact sheet. It is validated like any chat answer.
5. **If the narrative fails validation twice**, a rule-based narrative is shown instead, marked with a "Rule-based fallback" pill. The badges at the top always show validation status and confidence.

Recommended demo claim: CLM-011042. Cobalt Mutual, MRI brain (CPT 70553), billed $2,574.78, denied CO-197 (no authorization). I ran it today; the real output took 21.7 seconds (805 prompt + 128 generated tokens), passed validation on the first try, and returned:

- Finding (high): *CARC 197: Precertification/authorization/notification absent.*
- Confidence: high (0.85).
- Narrative (model-written): "*The claim was denied because the payer required prior authorization for this service, and no valid authorization was provided on the submitted claim… Verify whether an authorization was obtained but not reported; if so, submit a corrected claim with the authorization number.*"
- Actions: submit a corrected claim with the auth number if one exists → otherwise request retroactive authorization and appeal with medical records → add an auth check at scheduling for this payer and procedure.
- Sources: `denials_and_rejections.md`.

Say: "This one claim is the whole Cobalt story in miniature: a new authorization rule, a $2,500 MRI, and a fix that belongs at the front desk."

AI Copilot, Knowledge, Model and Settings pages

AI Copilot — what is on screen

A ChatGPT-style chat whose answers stream in word by word. Under every answer:

- **Badges:** a green "Validated" pill (or red "Validation failed"), an amber "Regenerated" pill if the first draft was rejected, the intent (Knowledge / Analytics / Claim / Data query), a "Grounded" pill, the time taken and the model name.
- **Data behind this answer** (opens by default for analytics): the exact numbers SQL computed and handed to the model. Point at this on stage — it is the proof the model did not invent the numbers.
- **Sources (n):** the knowledge passages used, with document, section and relevance score.
- **Validation details:** every check that fired, with severity.
- If validation corrects a streamed answer, the text is replaced in place and the reason shown.

How a Copilot question becomes a fact sheet. After the intent router picks "Analytics", pattern rules pick which SQL analysis to run:

If the question mentions…	SQL analysis run	Default window
prioritize / work first / which claims should	Ranks open claims by a priority score: balance, unresolved denial or rejection, recoverability, deadline risk and age	90 days
AR or Days in AR + increase / why / causing	AR drivers: AR change by payer, pending claims, adjudication speed	90 days
denial + increase / spike / why did	Denial drivers: rate now vs the previous period, by payer, provider and CARC	90 days
highest / lowest / worst / best + payer or provider	Ranking by denial rate (or days to adjudicate, AR, clean claim rate if mentioned)	at least 180 days
cause / reason / top + denial	Denial causes by CARC and category	at least 180 days
a claim ID such as CLM-011042	That claim's record and denial history	—
anything else analytical	KPI summary	90 days

Phrases such as "last 30 days" or "this year" change the window, and a payer or provider name in the question becomes a filter. Because rankings use at least 180 days, the Copilot's "Cobalt 26.5%" differs from the dashboard's 90-day 29.8% and the 12-month 17.0%. All three are correct for their windows.

RCM Knowledge page

- **Four cards:** documents (12), chunks (157), embedding model, and when the index was built.
- **Search box:** runs the same hybrid retrieval as the chat. Each hit shows three scores: the **relevance %** (cross-encoder, the one that decides ranking; green ≥ 50%), **dense** (meaning similarity from embeddings) and **bm25** (exact-word match). A hit can win on meaning, on exact words, or both.

- **Indexed documents table:** the 12 source files (ERA/EOB, denials and rejections, KPI definitions, prior authorization, payment posting, AR management, claims processing, coding, eligibility, patient responsibility, RCM workflow, curated Q&A). New `.md`, `.txt`, `.pdf` or `.docx` files dropped into `knowledge/` are added by re-running `scripts/ingest_knowledge.py`.
- **Chunking:** documents are cut into chunks of up to 1,000 characters, overlapping by 150 so a definition is never split mid-sentence without context.

Model page

Field	Value	What to say
Name	RCM-Qwen3-1.7B (QLoRA, Q4_K_M)	Our fine-tune of Alibaba's open Qwen3
File	rcm_qwen3_1.7b_q4_k_m.gguf, 1,056 MB	Fits on any laptop
Quantization	Q4_K_M	Weights stored in about 4 bits instead of 16 — about 4× smaller with little quality loss
Context	4,096 tokens used (trained up to 40,960)	Room for the question, references and fact sheet
Hardware	CPU only, 6 threads, 0 GPU layers	Load time 2.2 s
Thinking mode	Off	Qwen3's reasoning mode is disabled for speed

The page also shows live component status (LLM ready, RAG index ready, overall health) and a "How an answer is produced" summary of the pipeline.

Settings page

Backend URL and online/offline status, the dataset's as-of date and counts, the synthetic-data notice, and three Copilot preferences saved in the browser: ground answers in the knowledge base (on/off), temperature and maximum answer length.

Guardrails — what the validator actually checks

Check	Catches	Severity
Definitions	Wrong acronym expansion ("ERA = Electronic Revenue Attribution") or a known misdefinition ("an EOB is a bill", "appeal a rejection") for 15 critical terms	Error → regenerate
Missing core concept	A definition of ERA that never mentions remittance or 835	Error
KPI formulas	A stated formula missing required parts (Days in AR without "average daily")	Error
Numbers	Figures not found in the fact sheet or references	Warning/error
Direction	Saying a metric rose when the data says it fell	Error
Invented IDs	Claim numbers that appear in no source; masked if they survive the retry	Error
PHI	SSN-, phone-, email-, MRN- or date-of-birth-like content	Error
Repetition	A small model looping on the same sentence	Error / warning
Relevance	An answer that does not address the question	Error
Truncation, citations	Hit the token limit; cites a reference that does not exist; cites none	Warning / info

Off-topic questions ("write me a poem") and greetings get a fixed reply with no model call. A question with no RCM words is answered only if the knowledge base finds something relevant (cross-encoder score ≥ 0.2).

Other protections: rate limiting on the chat and analyze endpoints, input-length and pattern validation on every API parameter, CORS limited to the frontend's origin, secrets only in environment variables, and logs that record intent and timings but never the question text or patient data.

Demo script and likely questions

The demo works because the synthetic data contains four planted stories, and every chart and Copilot answer above traces back to them. Say so openly: it shows the analytics find real patterns rather than random noise.

The four stories planted in the data

ID	What was planted	Where it shows up
S1	Cobalt Mutual adds prior-authorization rules on 1 May 2026 (imaging, orthopedics, cardiology, PT); staff miss many	Denial rate rises from May; CO-197 tops the 90-day reasons; authorization = 41% of denied dollars
S2	Cobalt Mutual slows decisions from 1 July 2026 (about 48 days instead of 24)	Days in AR and outstanding AR step up in July; Cobalt has 223 pending claims
S3	Lakeside Dermatology keeps making modifier and bundling errors (CO-4, CO-97)	Joint-highest provider denial rate, 15.9%
S4	State Medicaid rejects more claims at the front door (member ID, eligibility)	Lowest clean claim rate (92.3%)

Before you present

1. Start the backend (`cd backend` , then `uvicorn app.main:app --host 127.0.0.1 --port 8000`) and the frontend (`cd frontend` , then `npm run dev`). Open http://localhost:3000.
2. Check the Model page shows "Loaded" and both components "Ready".
3. Ask one throwaway question first so the model is warm.
4. Answers take 15–25 seconds on CPU. Fill the wait by narrating the badges and the pipeline.

Ten-minute walkthrough

1. **Dashboard (2 min).** Read the headline. Point at Days in AR (53.9, red) and Outstanding AR (+36%). Walk charts 1 → 5 → 6: volume up a third, cash flat, AR and Days in AR climbing. End on Top denial reasons: CO-197 first.

2. **Analytics (1 min).** Payer table on 12M: Cobalt worst at 17.0%. Switch to 90D: 29.8%. "It's getting worse."
3. **Denials (1 min).** 74% preventable, 59.5% appeal win rate. Authorization: 17% of denials, 41% of dollars.
4. **Copilot (3 min).** Ask "What is an ERA?" — show Sources and the Validated badge. Then "Why did our denial rate increase?" — open *Data behind this answer* and say "SQL computed this; the model only explained it."
5. **Claim (2 min).** Open CLM-011042 (search it on the Claims page), click *Analyze with AI*. Talk through the panels while it runs, then the findings, confidence and actions.
6. **Model page (1 min).** "1 GB file, running on this laptop's CPU, no data leaves the machine."

What the Copilot actually answered today

I ran the demo questions against the live model on 9 Oct 2026. All four passed validation on the first draft. Wording varies slightly between runs, but the numbers come from SQL and stay the same.

Question	Time	What it said
Why did our denial rate increase?	20.6 s	11.7% → 12.8%. Cobalt Mutual went from 18.2% to 30.5% (+1.2 pts of the overall change); CARC 197 denials rose 47 → 75 (+1.4 pts)
What is causing our AR to increase?	23.6 s	AR $479,050 → $652,062 (+36.1%). Cobalt is the largest increase (+$69,016), and its decision time went from 28 to 47 days
Which claims should we prioritize?	21.4 s	Work CLM-007104, CLM-010480 and CLM-010407 first: high balances, unresolved, recoverable denials
What is the difference between a claim rejection and denial?	15.0 s	Rejection happens before payer processing (acknowledgment), is fixed and resubmitted; denial happens after adjudication (remittance) and may be appealed. 4 sources

Be ready for this one: "The dashboard says denial rate went *down* 0.6 pts, but the Copilot says it went *up*." Both are right. The Copilot deliberately leaves out claims from the last 30 days, because most of them have not been decided yet and would make the rate look artificially low. The dashboard card includes them. This is the "immature cohort" effect, and it is a good example of the analytics layer encoding RCM judgment the model could not.

Read answers before you present them. In today's denial answer, the second recommendation mixed up two codes ("authorization processes for claims with missing information (CARC 16)"). The validator does not catch every wording slip, only factual contradictions of definitions and data. If it happens live, point it out: it is the honest argument for keeping a human reviewer on recommended actions.

Questions you will probably get

"Why a small local model instead of GPT or Claude?" Claims data is PHI in real life, so keeping it on our own hardware avoids sending it to a third party. There is no per-question cost and it works offline. The architecture is the point: once the model only explains numbers and definitions it is handed, a 1.7B model is enough.

"If you have RAG, why fine-tune at all?" Fine-tuning taught the model RCM vocabulary, tone and answer structure. RAG supplies the facts, SQL supplies the numbers, and the validator enforces both. Each layer covers a weakness of the others.

"How do you know it isn't hallucinating?" Every answer is checked against the retrieved definitions, the SQL fact sheet and PHI patterns before it is shown. Open *Validation details* to show it. Failures are regenerated with stronger context; repeated failures fall back to the verified definition and say so.

"How accurate is it?" 103 automated backend tests pass, covering intent routing, retrieval, validators and analytics. Be honest that there is no recorded accuracy benchmark for the model itself yet.

"Is this real data?" No. It is fully synthetic, with fictional payers, providers and opaque patient IDs. The four planted scenarios make the patterns discoverable.

"Can it run on real data / at scale?" The schema is PostgreSQL-compatible and switches with one setting (`DATABASE_URL`); it currently runs on SQLite locally. Loading real 837/835 files would be the next integration step.

"Why is Payment Rate only 28%?" It is measured against billed (list-price) charges, and recent claims are not paid yet. Net Collection Rate (92.1%) is the fair measure.

"Why do three screens show three different Cobalt denial rates?" Different windows: 90 days on the dashboard (29.8%), 12 months on Analytics (17.0%), at least 180 days in the Copilot ranking. All three are correct for their window, and the gap itself shows the problem is recent.

What is not done yet — say it before someone asks

- Docker and deployment packaging (Phase 8) are not built.
- PostgreSQL is supported but has not been exercised; the demo runs on SQLite.
- The model rarely writes inline [1]-style citations. Sources are still attached to every answer.
- Answers take 15–25 seconds on CPU; a GPU build of llama.cpp would cut that sharply.
- The training data has known quality issues (documented in `training/README.md`): a large block of identical claim-review answers, and claim IDs in targets the model could not see. The runtime ID check and masking exist because of this. A retrain on cleaned data is the planned fix.